

MULTIPLE CHOICE QUESTIONS

1. A program loaded into memory and executing is called a:

- A. Process**
- B. Job
- C. Task
- D. Thread

2. The job pool is typically located on:

- A. Main memory
- B. Cache
- C. Hard disk**
- D. Registers

3. Job scheduling is used to:

- A. Choose which processes in pool to bring to main memory**
- B. Choose process in main memory to execute next
- C. Allocate CPU time to threads
- D. Manage I/O devices

4. CPU scheduling is needed when:

- A. Several jobs in main memory are ready to execute**
- B. A job is waiting for I/O
- C. The system boots
- D. A new job is added to the job pool

5. Multiprogramming organizes jobs so that:

- A. The CPU always has one to execute**
- B. Only one job runs at a time
- C. I/O devices are idle
- D. Users can interact directly

6. When waiting for I/O, the OS switches jobs using:

- A. Job scheduling
- B. CPU scheduling**
- C. Swapping
- D. Paging

7. Timesharing is a logical extension of:

- A. Multiprogramming**
- B. Multitasking
- C. Multiprocessing
- D. Networking

8. In timesharing, response time should be:

- A. < 1 second**
- B. < 1 millisecond
- C. > 1 second
- D. > 1 minute

9. Swapping is used when:

- A. Processes don't fit in memory**
- B. The CPU is idle
- C. I/O is completed
- D. A job is created

10. Virtual memory allows:

- A. Execution of processes not completely in memory**
- B. Faster disk access
- C. More registers
- D. Larger cache

11. Modern operating systems are:

- A. Interrupt driven**
- B. Polling driven

12. A software interrupt caused by error or request is:

- A. Trap (exception)**

Operating Systems

CHAPTER 1: REVIEW & SELF-ASSESSMENT (PAGES 24–30)

Part 2: Multiple Choice (17–35)

17. Privileged instructions can only be executed in:

- A. Kernel mode**
- B. User mode
- C. Any mode
- D. Debug mode

18. A system call changes the mode to:

- A. Kernel mode**
- B. User mode
- C. Idle mode
- D. Safe mode

19. To prevent infinite loops, the OS uses:

- A. A timer**
- B. A cache
- C. A semaphore
- D. A mutex

20. The timer generates an interrupt after:

- A. A specific period**
- B. Each instruction
- C. Each I/O operation
- D. Each system call

21. A process is:

- A. A program in execution**
- B. A passive entity
- C. A file on disk
- D. A hardware component

22. Which is NOT a resource needed by a process?

- A. CPU
- B. Memory
- C. I/O
- D. Compiler**

23. Process termination requires:

- A. Reclaim of any reusable resources**
- B. Deletion of all files
- C. Formatting the disk
- D. Restarting the system

24. A single-threaded process has:

- A. One program counter**
- B. Multiple program counters
- C. No program counter
- D. One stack

25. A multi-threaded process has:

- A. One program counter per thread**
- B. One program counter for all threads
- C. No program counter
- D. One PC per process

26. Concurrency in multi-process systems is via:

- A. Multiplexing CPUs among processes**
- B. Sequential process execution
- C. Using multiple disks
- D. Increasing memory

27. Which is NOT a process management activity?

- A. Creating & deleting processes
- B. Suspending & resuming processes

28. OS provides mechanisms for:

- A. Process communication
- B. Deadlock handling
- C. Both A and B**

ESSAY & CONCEPTUAL QUESTIONS

1. Explain the concepts of multiprogramming and timesharing (multitasking). How do they differ, and what is the role of job scheduling and CPU scheduling in each?

Answer: Multiprogramming organizes jobs (code and data) so that the CPU always has one to execute. A subset of total jobs is kept in memory. When a job has to wait (e.g., for I/O), the OS switches to another job using CPU scheduling. This increases CPU utilization.

Timesharing is a logical extension of multiprogramming in which the CPU switches jobs so frequently that users can interact with each job while it is running, creating interactive computing. Response time should be < 1 second. Each user has at least one program executing in memory (process). If several jobs are ready to run, CPU scheduling is used. If processes don't fit in memory, swapping moves them in and out. Virtual memory allows execution of processes not completely in memory. A timeshared OS uses both CPU scheduling and job scheduling to provide users with a small portion of a time-shared computer.

2. Describe dual-mode operation in operating systems. Why is it necessary? Explain the mode bit, user mode, kernel mode, privileged instructions, and how system calls, traps, and interrupts cause mode changes.

Answer: Dual-mode operation allows the OS to protect itself and other system components. It uses two modes: **user mode** and **kernel mode** (also called supervisor, privileged, or system mode). A mode bit provided by hardware distinguishes between them: 0 for kernel mode, 1 for user mode. Some instructions are designated as **privileged** and can only be executed in kernel mode. System calls, traps, and interrupts change the mode to kernel. A system call is issued when a user application requests a service from the OS. This transition ensures that the OS maintains control over the CPU and prevents user programs from executing privileged operations directly.

3. Explain the purpose of the timer in an operating system and how it helps maintain control over the CPU.

Answer: To ensure that the OS maintains control over the CPU, it uses a **timer**. The timer is set to interrupt the computer after a specific period. The operating system decrements a counter until it reaches zero; when the counter is zero, an interrupt is generated. The timer is set up before scheduling a process to regain control or terminate a program that exceeds its allocated time. This prevents infinite loops or a process from hogging resources indefinitely.

4. Discuss the process management responsibilities of an operating system. What is a process, what resources does it need, and what activities does the OS perform in connection with process management?